Medofile

COMPLETE PRODUCT AND ENGINEERING DOSSIER

Connected healthcare for patients and doctors.

A complete overview of Medofile: product purpose, user workflows, AI services, technology stack, system architecture, data model, security, SEO, deployment, and current implementation status.

Production website
https://medofile.com

Platform
Web + Expo mobile + Express API

Audience
Patients, doctors, clinics, administrators

Prepared from the current Medofile implementation and project configuration.

EXECUTIVE SUMMARY

What Medofile is

Medofile is a connected healthcare platform that helps people find trusted doctors, make informed booking decisions, and stay organized before and after appointments.

> **Core product idea:** Medofile is not only a doctor booking directory. It connects discovery, booking, reminders, follow-up instructions, care history, verified reviews, health education, and AI-guided next steps in one care journey.

Discover

Search doctors by specialty, city, symptoms, availability, fee, language, consultation type, or current location.

Decide

Compare doctor profiles, experience, fees, ratings, treatments, reviews, and care options.

Continue care

Manage appointments, receive reminders, view timelines, read follow-up instructions, and keep care information organized.

Product goals

- Make healthcare discovery easier and more local.
- Give patients more confidence before booking.
- Help doctors build a trustworthy digital presence.
- Support patients after an appointment, not only before it.
- Use AI for guidance and organization while keeping doctors responsible for diagnosis and treatment.

02 · PEOPLE AND PERMISSIONS

Users and roles

Role	Primary purpose	Key capabilities	Authentication
Guest patient	Browse and book without an account.	Search doctors, view profiles, submit guest bookings.	Not required for browsing or guest booking.
Patient	Manage a personal care journey.	Bookings, saved doctors, cancellation, rescheduling, care timeline, reminders, follow-ups, verified reviews, AI assistant.	Clerk patient account.
Doctor	Manage professional presence and patient leads.	Registration, profile editing, specialties, availability, bookings, patient view, follow-up instructions, practice insights.	Clerk doctor account linked to doctor profile.
Admin	Operate and moderate the platform.	Doctors, bookings, blog, settings, analytics, users, moderation.	Separate JWT authentication.
Superadmin	Manage privileged administration.	All admin functions plus superadmin-protected actions.	JWT with role validation.

Identity model

Clerk remains the identity provider. PostgreSQL stores only the application data Medofile needs, linked to the Clerk user ID. This creates a clear ownership chain for patients, doctors, saved doctors, bookings, care events, follow-ups, and reviews.

```
Clerk user ID
    ↓
patientProfiles.clerkUserId / doctors.ownerClerkUserId
    ↓
bookings.patientClerkUserId
    ↓
careTimelineEvents.patientClerkUserId
    ↓
reminders, documents, follow-ups, saved doctors
```

03 · PATIENT EXPERIENCE

Complete patient workflow

1. Discover Visit Medofile or search online. → **2. Search** Use specialty, city, symptom, or location. → **3. Compare** Review doctor profiles and care options. → **4. Book** Choose date, time, and consultation type. → **5. Continue** Receive reminders and follow-up care.

1. Discover care

The homepage is optimized for intent such as “doctor near me”, “find a trusted doctor”, “doctor by specialty”, and “book a doctor appointment”. Patients can browse before signing in.

2. Search and location matching

The search screen supports doctor name, specialty, city, language, gender, consultation type, availability, fee, rating, experience, and current location. When location permission is granted, the browser sends coordinates to the API and doctors are filtered and ranked by distance.

3. Compare doctors

A profile presents qualification, experience, specialty, fees, rating, review count, clinic and hospital information, languages, timings, treatments, education, memberships, insurance, branches, social profiles, and booking availability.

4. Book an appointment

The booking endpoint validates the doctor, patient contact details, date, time, and care type. Signed-in bookings are associated with the patient’s Clerk ID. Guest bookings remain supported.

5. Manage appointments

Signed-in patients can see appointment status, cancel eligible upcoming bookings, or reschedule them. Each important action creates a care timeline event so the patient has a record of what changed.

6. Continue after the appointment

Doctors can add follow-up instructions. Patients can see reminders, appointment events, follow-ups, and care history in the care space.

7. Review after completed care

Reviews require a completed appointment with the doctor. Suspicious or inappropriate content can be held for moderation, and published verified reviews update doctor ratings.

04 · DOCTOR EXPERIENCE

Doctor and practice workflow

Doctor onboarding

1. Doctor creates a Clerk account.
2. Doctor opens the registration flow.
3. Doctor provides professional identity and contact details.
4. Doctor selects one or more specialties.
5. Doctor adds clinic, city, timing, fees, and consultation types.
6. Doctor submits the profile for platform use and potential verification.

Profile presentation

- Specialty and qualification
- Experience and ratings
- Clinic and hospital details
- Languages and treatments
- Education and memberships
- Insurance and branches
- Availability and booking options
- Location and social links

Doctor dashboard

The doctor dashboard centralizes booking leads, patient appointments, profile editing, follow-up instructions, and practice insights. Analytics include impressions, profile views, calls, chats, bookings, directions, and weekly trends.

Specialty storage decision

Doctor specialties currently remain in the existing string field as comma-separated values. This preserves backward compatibility with older doctor records while allowing the web registration and profile editor to use a searchable multi-select UI.

05 · ADMIN OPERATIONS

Administration workflow

Authenticate
Admin signs in with JWT credentials.

→

Operate
Manage doctors, bookings, users, and content.

→

Moderate
Review doctor profiles, reviews, and blog content.

→

Measure
Review analytics and platform activity.

Admin authentication

Admin credentials are validated server-side. Passwords are bcrypt-hashed and successful login produces a signed JWT. Protected API routes validate the Bearer token. Superadmin endpoints additionally require the `superadmin` role.

Admin responsibilities

- Maintain doctor profile quality and verification state.
- Review bookings and operational activity.
- Moderate suspicious or pending reviews.
- Manage blog settings and generated health articles.
- Monitor platform analytics and content performance.
- Manage authorized administrator accounts.

06 · AI SERVICES

AI agents and automated intelligence

Important distinction: The production product uses controlled AI services for specific healthcare and content tasks. It does not run an autonomous general-purpose coding agent in the live application. Replit Agent assisted the project's development, but the runtime AI features are Medofile services.

AI or automation	Technology	What it does	Safety or control
Medofile AI Health Assistant	OpenAI-compatible SDK through Replit AI Integration; `gpt-5-mini`; SSE streaming.	Discusses symptoms, suggests specialist types, explains urgency, provides general wellness guidance, and returns structured doctor suggestion tags.	No definitive diagnosis, no dosage advice, doctor recommendation, emergency guidance for serious symptoms.
Prescription Image Analyzer	Vision-capable `gpt-5-mini` request through `/api/chat/prescription`.	Extracts readable medicine names, general categories, and visible instructions.	Does not diagnose, prescribe, change medication, or invent unreadable instructions.
Daily Health Blog Generator	OpenAI-compatible SDK; structured JSON; hourly scheduler check.	Creates patient-friendly health guides from approved topics, including SEO descriptions and related specialties.	Required disclaimer, prohibited-language validation, dosage rejection, similarity detection, draft/publish control.
Doctor Match Engine	Deterministic TypeScript scoring; not an LLM.	Ranks doctors using specialty, city, distance, rating, language, fees, consultation type, availability, treatments, and profile relevance.	Explainable reasons are returned with every match.
Reminder Scheduler	Database-backed timed service; not AI.	Creates and delivers due appointment reminders into the patient care timeline.	Checks appointment status and patient in-app reminder preference before delivery.

AI Health Assistant workflow

Patient message Symptoms or question. → **Safety prompt** Rules and location context. → **AI response** Streamed through SSE. → **Doctor tag** Structured specialties and city. → **Matching API** Relevant Medofile doctors.

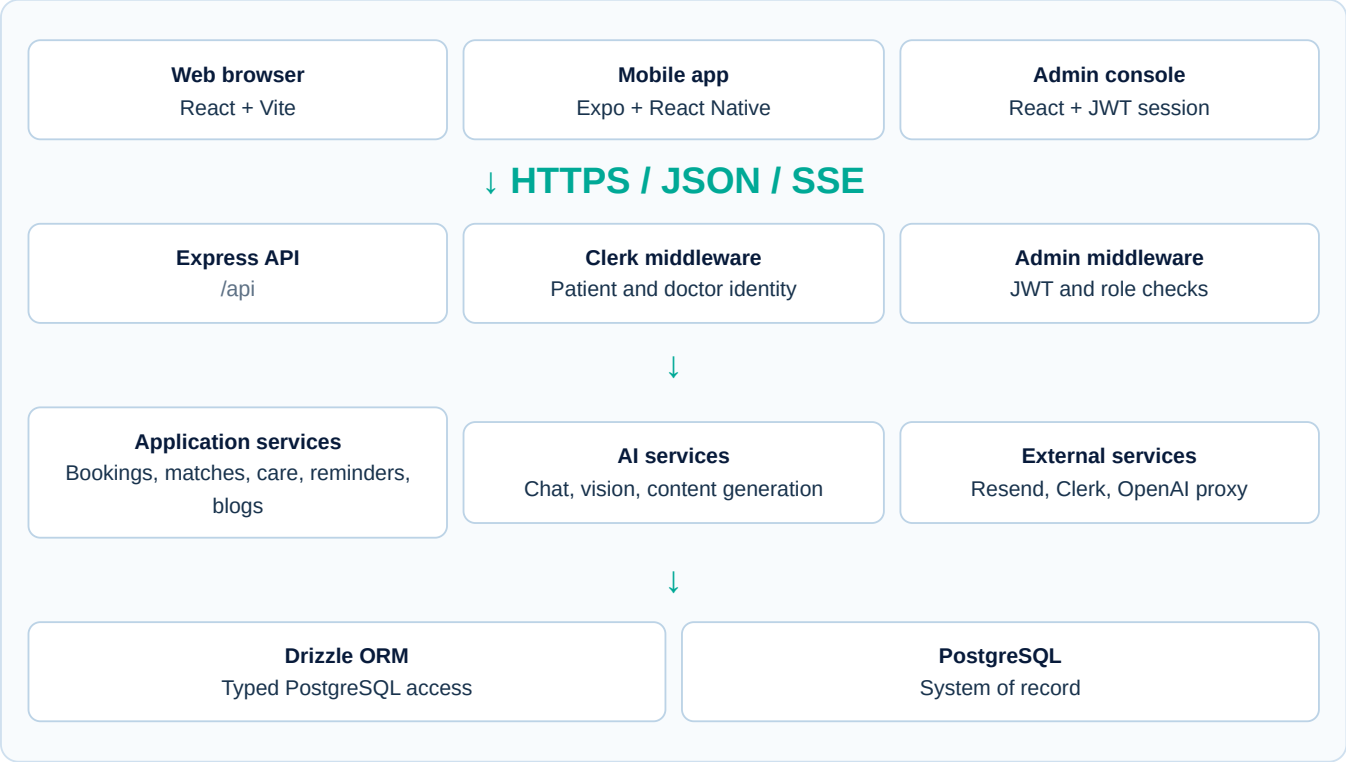
07 · TECHNOLOGY

Technology stack

Layer	Technology	Responsibility
Web	React, TypeScript, Vite	Patient, doctor, admin, blog, legal, contact, and public pages.
Routing	Wouter	Client-side route handling for public, authenticated, doctor, and admin experiences.
Frontend data	TanStack React Query, fetch, generated API client libraries	Server state, caching, loading, mutation, and API communication.
UI	Tailwind CSS, Radix UI, shadcn-style components, Lucide React	Accessible interface primitives, responsive layout, icons, forms, dialogs, menus, and cards.
Animation	Framer Motion	Homepage motion, transitions, doctor cards, and interactive visual elements.
Mobile	Expo SDK, React Native, Expo Router, TypeScript	Patient and doctor mobile application flows.
Backend	Node.js, Express 5, TypeScript	REST API, authentication middleware, AI endpoints, booking logic, content services, and schedulers.
Database	PostgreSQL, Drizzle ORM	Persistent source of truth for doctors, bookings, care, reviews, reminders, blogs, and admin data.
Validation	Zod, Drizzle-Zod, API schemas, OpenAPI/Orval tooling	Boundary validation and shared API type generation.
Authentication	Clerk for patients/doctors; JWT and bcrypt for admins	Account identity, sessions, route authorization, and role enforcement.
AI	OpenAI-compatible SDK through Replit AI Integrations	Chat, prescription vision, and daily health content generation.
Email	Resend through Replit Connectors SDK	Server-side contact enquiry delivery to admin@digitaldhatu.com.
Logging	Pino and pino-http	Structured backend request and service logging.
Deployment	Replit artifact workflows	Runs web, API, mobile, overview, and component preview services.

08 · SYSTEM ARCHITECTURE

How the platform works technically



Standard request flow

```
Browser or mobile app
        ↓
React / React Native screen
        ↓
React Query or fetch request
        ↓
Express API under /api
        ↓
Authentication and authorization middleware
        ↓
Route handler and validation
        ↓
Drizzle ORM
        ↓
PostgreSQL
        ↓
JSON or Server-Sent Events response
        ↓
UI state and visible result
```

AI request flow

```
Patient message or prescription image
        ↓
/api/chat or /api/chat/prescription
        ↓
Medofile safety prompt and request shaping
        ↓
OpenAI-compatible Replit AI Integration
```

```
        ↓
SSE or JSON response
        ↓
Client-side rendering and structured doctor search
```

09 · DATABASE

Data model and source of truth

PostgreSQL is the source of truth for operational Medofile data. The frontend may cache responses, but core records are created and updated through the API.

Data group	Models	Purpose
Doctors	doctors	Professional profile, location, specialty, practice information, availability, and analytics.
Accounts	patientProfiles, savedDoctors	Application-owned patient profile projection and saved doctor relationships.
Appointments	bookings	Patient contact, doctor, time, fee, consultation type, status, cancellation, and rescheduling.
Care history	careTimelineEvents, careDocuments	Timeline events and patient care documents.
Follow-up	doctorFollowUps	Doctor instructions associated with an appointment and patient.
Trust	doctorReviews	Verified appointment reviews, moderation state, and ratings.
Reminders	reminderPreferences, scheduledReminders	Reminder channel preferences, due times, delivery state, and timeline delivery.
AI chat	conversations, messages	Conversation and message storage for supported chat experiences.
Content	blogArticles, blogSettings, blogGenerationLogs, blogGenerationRuns	Published/draft articles, generation settings, locks, logs, and daily generation state.
Administration	adminUsers	Admin credentials, role, and access management.

10 · API WORKFLOW

API route groups

Route group	Examples	Function
Health and platform	`/api/health`	Service availability and health checks.
AI	`/api/chat`, `/api/chat/prescription`	Streaming assistant responses and prescription vision extraction.
Doctors	`/api/doctors`, `/api/doctors/:id`, `/api/doctors/register`	Search, profiles, registration, editing, verification, analytics, and reviews.
Matching	`/api/doctor-matches`, `/api/doctors/suggestions`	Explainable doctor ranking and specialty/city suggestions.
Bookings	`/api/bookings`, `/api/me/bookings`	Create, list, update, cancel, and reschedule appointments.
Accounts	`/api/me/*`	Patient profile, saved doctors, care history, reminders, and doctor-owned data.
Blog	`/api/blog`, admin blog routes	Read content, generate articles, manage settings, and moderate drafts.
Contact	`/api/contact`	Validate enquiries and send them server-side through Resend.
Admin	JWT-protected admin routes	Operations, moderation, content, users, and platform management.

11 · SECURITY AND PRIVACY

Security model

Identity separation

Clerk manages patient and doctor identity. Admin users use a separate JWT system. This prevents mixing public accounts with internal administration.

Ownership scoping

Patient bookings, saved doctors, care events, follow-ups, and reminders are associated with the relevant Clerk user ID.

Server-side secrets

OpenAI, Clerk secret, session, JWT, and Resend integration credentials are used server-side and are not shipped to the browser.

AI guardrails

Medical responses avoid definitive diagnosis, dosage advice, and medication changes. Serious symptoms receive emergency guidance.

Review protection

Reviews are tied to completed appointments and validated for length, contact details, links, and suspicious language.

Contact protection

Contact enquiries use input limits, email validation, a honeypot field, rate limiting, and generic delivery errors.

Healthcare boundary: Medofile provides discovery tools, educational information, and AI-assisted guidance. It does not replace a qualified healthcare professional or emergency services.

12 · SEO AND GROWTH

Search visibility strategy

Medofile is optimized around doctor discovery intent, especially “doctor near me” searches in India.

On-page SEO

- Homepage title targets “Doctor Near Me in India”.
- Homepage H1 includes “Find a trusted doctor near me”.
- Doctor listing page has a unique title and description.
- Doctor profiles receive dynamic metadata.
- Supporting copy explains specialty, city, and location search.

Technical SEO

- Canonical URLs
- Open Graph and Twitter metadata
- Organization and WebSite structured data
- SearchAction schema
- robots.txt
- sitemap.xml
- Descriptive internal links

Important ranking expectation

SEO implementation makes the site understandable and crawlable, but rankings depend on search-engine crawling, indexing, location relevance, site authority, page quality, competition, and ongoing content publishing. Google Search Console should be used to monitor actual impressions and queries.

13 · DEPLOYMENT

Runtime and deployment workflow

Develop
Work in the web, mobile, API, and shared packages.

→

Validate
Typecheck, build, inspect logs, and preview.

→

Run workflows
Web, API, mobile, overview, and sandbox.

→

Publish
Deploy the verified artifacts to Medofile's domain.

Artifact	Role	Development workflow
`artifacts/web`	Public web application	Vite development server and production static build.
`artifacts/api-server`	Express API	Build with esbuild, run Node server on port 8080.
`artifacts/mobile`	Expo mobile app	Expo and Metro development workflow.
`artifacts/medofile-overview`	Project overview presentation	Vite-based overview artifact.
`artifacts/mockup-sandbox`	Component preview environment	Isolated Vite preview workflow for design exploration.

Production website: https://medofile.com

14 · CURRENT STATUS

Implemented, partial, and next-stage work

Status	Area	Details
Implemented	Patient platform	Search, location matching, profiles, guest booking, accounts, saved doctors, appointment management, care timeline, follow-ups, reminders, reviews, AI assistant, prescription analysis.
Implemented	Doctor platform	Registration, searchable multi-specialty selection on web, profile editing, bookings, patient view, follow-ups, analytics, image handling.
Implemented	Admin platform	Separate JWT authentication, role protection, doctor and booking operations, blog management, content controls, analytics.
Implemented	Public website	Home, doctor discovery, blog, About, Careers, Contact, Privacy Policy, Terms, shared footer, SEO metadata, sitemap, structured data.
Implemented	AI services	Streaming symptom assistant, prescription image extraction, automated health blog generation with validation and deduplication.
Expanding	Secure attachments	Prescription and health-report attachment support is being expanded for persistent, secure patient documents.
Expanding	Mobile parity	Mobile multi-specialty selection and secure document attachment workflows are being brought to parity with the web.
Expanding	Reminder channels	Preference fields exist for email, push, and WhatsApp; the active scheduler currently delivers in-app timeline reminders.
Design decision	Specialty storage	Specialties remain comma-separated in the existing field for backward compatibility while the UI supports multi-select editing.

15 · END-TO-END EXAMPLES

How common scenarios work

Example A: Patient searches for a cardiologist near Mumbai

1. Patient opens Medofile and searches for cardiology.
2. Patient selects Mumbai or allows location access.
3. Frontend requests doctors with specialization, city, latitude, longitude, and radius.
4. API filters and sorts relevant doctors.
5. Doctor cards show ratings, experience, fees, clinic details, and availability.
6. Patient opens a profile and requests a booking.
7. API creates the booking and, if signed in, adds the event to the care timeline.

Example B: Patient asks the AI about recurring headaches

1. Patient sends the symptom through the AI Assistant.
2. API adds the safety system prompt.
3. AI describes several possibilities without diagnosing.
4. AI suggests an appropriate specialist, such as a neurologist or general physician.
5. AI returns a structured doctor recommendation tag.
6. Frontend requests matching doctors.
7. Patient can open a doctor profile and book care.

Example C: Doctor adds post-appointment instructions

1. Doctor opens a completed patient booking.
2. Doctor writes follow-up instructions.
3. API verifies the doctor and booking relationship.
4. Follow-up is stored against the appointment and patient Clerk ID.
5. A care timeline event is created.
6. Patient sees the instruction in the care section.

Example D: Visitor sends a contact enquiry

1. Visitor completes the contact form.
2. Frontend sends the form to `/api/contact`.
3. API validates fields, checks the honeypot, and applies rate limiting.
4. Resend delivers the enquiry to **admin@digitaldhatu.com**.
5. The visitor receives a success or friendly error message.

16 · REFERENCE

Key project locations

Location	Purpose
`artifacts/web/src/App.tsx`	Web routing, Clerk provider, shared footer, floating assistant, and application shell.
`artifacts/web/src/pages/`	Patient, doctor, admin, blog, legal, public, and account screens.
`artifacts/web/src/lib/seo.ts`	Route-aware page titles, descriptions, Open Graph tags, Twitter tags, and canonical URLs.
`artifacts/api-server/src/routes/`	Express API route groups for chat, doctors, bookings, accounts, blogs, contact, and admin operations.
`artifacts/api-server/src/services/`	Reminder scheduling and AI blog publishing services.
`lib/db/src/schema/`	Typed Drizzle PostgreSQL schema.
`lib/integrations-openai-ai-server/`	Server-side OpenAI-compatible integration client and streaming helpers.
`artifacts/mobile/app/`	Expo Router mobile screens for patients, doctors, booking, chat, and healthbot.
`artifacts/web/public/robots.txt`	Crawler access policy and sitemap reference.
`artifacts/web/public/sitemap.xml`	Public route index for search engines.

One-line summary: Medofile connects doctor discovery, location-aware matching, booking, AI-assisted guidance, and post-appointment care into one patient and doctor platform.